

QueueFlow — Project Structure & Architecture Guide

Comprehensive Technical Structure & Judge Reference Document | Built by LahootiX

System Summary for Judges: QueueFlow is a full-stack digital queue and appointment booking platform built with Next.js 15 App Router, Supabase PostgreSQL with transactional RPCs & Realtime WebSockets, and Google Gemini 1.5 Flash AI. All database operations use transactional advisory locks and Row-Level Security.

1. High-Level Architecture & Connectivity

QueueFlow connects five core architectural layers into an automated, event-driven ecosystem:

- Frontend Client Layer (Next.js 15 App Router):** Role-specific web interfaces for Customers (/book, /token, /my), Staff (/staff), Managers (/manager), Admins (/admin), and a public TV display kiosk (/display).
- Backend & Server Action Layer (Node.js):** Server Actions (lib/actions/*) execute server-side business workflows with strict Zod validation and session authorization before reaching the database.
- Database & Stored Procedures (Supabase PostgreSQL):** Single source of truth (supabase/functions.sql). Transactional RPCs handle concurrency, ticket generation, slot validation, and status transitions with zero race conditions.
- Realtime Synchronization Engine (WebSockets):** Supabase Realtime broadcast channels notify active customer views (/my) and display boards (/display) the moment a staff member calls a token, without manual page refreshing.
- AI & Intelligence Services (Gemini + Algorithms):** Google Gemini 1.5 Flash generates actionable managerial operations insights, powers the interactive live assistant widget, and algorithmic heuristics predict wait times and no-show risks.

2. Important Folders & Their Purpose

Folder / Directory	Filesystem Path	Architectural Purpose
app/	app/	Next.js 15 App Router root containing all role pages, layouts, and API routes.
app/book/	app/book/	Interactive multi-step appointment booking wizard with real-time slot selection.
app/token/	app/token/	Walk-in ticket generation interface with live estimated wait time display.
app/my/	app/my/	Customer personal portal displaying active live token card and visit

		history.
app/staff/	app/staff/	Staff counter terminal supporting Call Next, Start, Complete, Skip, and Recall.
app/manager/	app/manager/	Operations dashboard with Recharts analytics, KPI cards, and AI staffing tips.
app/admin/	app/admin/	System governance panel managing rules, services, departments, and user roles.
app/display/	app/display/	Full-screen live TV queue board for waiting area kiosks.
app/api/ai/	app/api/ai/	Server-side Gemini AI endpoints for managerial insights and live assistant.
components/	components/	Reusable UI widgets: AI Assistant Widget, AI Insights Card, Notification Bell.
lib/actions/	lib/actions/	Server Actions with Zod schemas for appointments, tokens, staff, and admin.
lib/ai/	lib/ai/	Heuristic AI modules for wait-time prediction, no-show scoring, and staffing.
lib/supabase/	lib/supabase/	SSR-compatible Supabase database client initializers.
supabase/	supabase/	SQL migrations: schema.sql (RLS & tables), functions.sql (RPCs), seed.sql.

3. Important Files & Key Implementation Responsibilities

File Name	Relative Location	Key Implementation Responsibility
functions.sql	supabase/functions.sql	Core database logic: transactional RPCs with advisory locks for booking, queueing, and status transitions.
schema.sql	supabase/schema.sql	Database schema definition: 9 tables, relational foreign keys, and complete RLS security policies.
seed.sql	supabase/seed.sql	Seed dataset: 3 departments, 7 services, 5 counters, 6 demo accounts, and 14-day realistic queue history.
proxy.ts	proxy.ts	Next.js Edge middleware enforcing session validation and route-level role-based access control.
appointments.ts	lib/actions/appointments.ts	Server actions executing appointment booking, cancellation, rescheduling, and check-in.

tokens.ts	lib/actions/tokens.ts	Server actions generating walk-in tokens and checking queue limits.
staff.ts	lib/actions/staff.ts	Server actions for staff counter control (Call Next, Start, Complete, Skip, Status).
admin.ts	lib/actions/admin.ts	Server actions for administrative configuration (rules, user roles, working hours).
waitTime.ts	lib/ai/waitTime.ts	Wait time prediction model factoring queue depth, service averages, and 11am-1pm peak surge.
noShow.ts	lib/ai/noShow.ts	No-show risk scoring algorithm (0.0 to 1.0) assessing lead time, slot time, and customer history.
staffRecommendation.ts	lib/ai/staffRecommendation.ts	Dynamic staffing formula: $\text{ceil}(\text{demand} * \text{avg_service_duration} / 60)$ for counter capacity.
insights/route.ts	app/api/ai/insights/route.ts	Gemini 1.5 Flash operations analyst endpoint with Zod validation and 5-min caching.
assistant/route.ts	app/api/ai/assistant/route.ts	Gemini 1.5 Flash conversational assistant backend for user help and system navigation.
ai-assistant-widget.tsx	components/ai-assistant-widget.tsx	Floating chat support widget accessible globally on all pages with pre-built prompt chips.

4. Implementation Blueprint: Appointment & Queue Logic

Where Appointment Logic Lives:

1. Slot Discovery: supabase/functions.sql -> get_available_slots() dynamically generates slot grids based on department working hours and designated break times, checking booked capacity in real time.
2. Booking Concurrency: supabase/functions.sql -> book_appointment() uses PostgreSQL advisory locks to guarantee zero overbooking under high concurrent load, enforcing the daily booking ceiling.
3. Check-In & Priority Tokens: supabase/functions.sql -> check_in() verifies that the customer is within the allowed arrival window (-30 min to +15 min) and generates an expedited Priority Token (prefix 'P-') placed at the head of the queue.
4. Missed Appointment Cleanup: supabase/functions.sql -> mark_missed_appointments() automatically transitions unfulfilled past slots to 'missed' status and releases counter capacity.

Where Queue & Token Logic Lives:

1. Walk-in Token Creation: supabase/functions.sql -> create_token() verifies daily token limits, prevents duplicate active tokens for the same user, and assigns sequential departmental tokens (e.g., A-001).
2. Dynamic Queue Recalculation: supabase/functions.sql -> recalc_queue() recalculates position order and wait times whenever a customer checks in or a counter changes status.

3. Counter Service Flow: supabase/functions.sql -> call_next_token(), start_service(), complete_service(), skip_token() execute atomic state transitions and notify connected clients over Supabase Realtime.

5. Implementation Blueprint: AI & Intelligent Features

Intelligent Feature	Code Location	Mechanism & Algorithmic Design
Predictive Wait Times	lib/ai/waitTime.ts app/token/page.tsx	Calculates ETA using queue depth, active counters, and historical duration averages, applying a 1.25x multiplier during peak hours (11:00 AM - 1:00 PM).
No-Show Risk Scoring	lib/ai/noShow.ts app/staff/page.tsx	Evaluates lead time, appointment hour, and past attendance to assign a 0.0-1.0 risk score, displayed as colored badges on the staff counter list.
Smart Staff Recommendation	lib/ai/staffRecommendation.ts app/manager/page.tsx	Uses operations research formula $\text{ceil}(\text{demand} * \text{duration} / 60)$ to recommend counter staffing levels to managers.
Manager Operations Insights	app/api/ai/insights/route.ts components/ai-insights-card.tsx	Google Gemini 1.5 Flash analyzes aggregated non-PII metrics to generate 3-5 concise, actionable operational recommendations.
In-App AI Assistant	app/api/ai/assistant/route.ts components/ai-assistant-widget.tsx	Conversational AI widget answering customer queries regarding appointment booking, queue status, and system navigation.

6. Security & Governance Architecture

- **Row-Level Security (RLS):** Active on all 9 tables. Customers only view their personal records; staff and managers view department queues; admins view all.
- **Role Escalation Protection:** PostgreSQL trigger prevent_self_role_escalation prevents non-admin users from altering their own role in the database.
- **Stored Procedure Hardening:** All security definer RPC functions explicitly specify SET search_path = public to neutralize search-path escalation attacks.
- **HTTP Security Headers:** next.config.ts enforces X-Frame-Options: DENY, X-Content-Type-Options: nosniff, Referrer-Policy, and Permissions-Policy.
- **Credential Protection:** Demo credentials and API keys are strictly kept server-side in .env.local; zero secrets are exposed in the client or committed to Git.

7. Demo Accounts & Verification Guide for Judges

Role	Login Email	Password	Primary Routes	Key Demo Action
Customer	customer@demo.com	Demo@12345	/book, /token, /my	Book appointment slot;

generate walk-in token;
track real-time position.

Customer 2	customer2@demo.com	Demo@12345	/book, /token, /my	Test concurrent queue progression and multi-user ordering.
Staff	staff@demo.com	Demo@12345	/staff	Select counter; call next token; view no-show risk badge; complete service.
Manager	manager@demo.com	Demo@12345	/manager	Review Recharts metrics, Gemini operational insights, and staff recommendations.
Admin	admin@demo.com	Demo@12345	/admin	Adjust queue limits, manage departments/services, and assign user roles.
Public Kiosk	(Public access)	N/A	/display	View full-screen live waiting room TV display with real-time token calling.